Sakif Hussain Shachcha

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Bengali(Native) | English(C2)

Education & Certificates

Master of Computer Science in Software Engineering **03/2026 – Present**
University of Wollongong, Australia

B.Sc. in Computer Science and Engineering **01/2020 – 04/2024**
Ahsanullah University of Science and Technology, Bangladesh

Technical Skills

- **Languages:** Python, Go, C++, TypeScript
- **Frontend:** React.js, HTMX, Alpine.js, Templ, Tailwind CSS
- **Backend & Data:** PostgreSQL, Redis, RabbitMQ, SQLite
- **Infrastructure & AI**: Docker, CrewAI, LangGraph, TensorFlow

Work History

Junior AI Engineer, SForger, Poland(Remote) **05/2025 – 03/2026**

- Re-architected legacy monolithic infrastructure into decoupled, Dockerized microservices using TypeScript, enabling seamless AI chatbot integration via RESTful APIs.
- Engineered a multi-agent system using CrewAI and Python, delivering autonomous integration tooling that allowed enterprise clients to consolidate business operations.

Junior Software Engineer, W3 Engineers LTD, Bangladesh **02/2025 – 05/2025**

- Built and deployed a business intelligence module using Python, React.js, and PostgreSQL, leveraging LangGraph to accelerate ad operations for the sales team.
- Developed an internal project documentation platform utilizing RAG architectures and VectorDBs, directly integrating with company codebases to reduce team onboarding time and documentation overhead.

Software Engineering Intern, W3 Engineers LTD, Bangladesh **11/2024 – 02/2025**

- Collaborated with senior engineering leadership to ship live production features using Django and Python, actively refining system architecture through peer code reviews.
- Built an automated travel itinerary-generation service using FastAPI and Python, transforming manual property listing forms into dynamic, user-ready content.
- Designed a PostgreSQL and Django analytics dashboard to aggregate Google and Meta ad performance metrics, reducing workflow complexity and improving observability for the marketing department.

Publication

Comparative Study of Region-Based Approaches to Road Image Classification of National Highways of Bangladesh Using Deep Learning Models - 2024 IEEE International Conference on Artificial Intelligence in Engineering and Technology (IICAIET)
Link:ieeexplore.ieee.org/document/10730626

- Developed a hybrid deep learning model utilizing covered area analysis to process and classify segmented road images via satellite imagery.
- Engineered the data preprocessing pipeline to handle and segment high-resolution highway imagery, optimizing the model to perform reliably in data-constrained environments.
- Achieved 98% accuracy in classifying curved and complex road conditions, demonstrating the practical application of computer vision for real-world infrastructure.

Projects

Tellproof — Open-Source AI Telemetry & Governance Framework
Project Link: `github.com/Lumetrix-Advanced-Technologies/tellproof`
Registry Links: `npmjs.com/package/tellproof` & `pypi.org/project/tellproof`

- Engineered a tamper-evident Go telemetry agent that aggregates LLM operations into hourly batches, utilizing salted Merkle tree commitments and Ed25519 signatures for verifiable audit logs.
- Developed Python and Node.js SDKs to instrument major AI providers via OpenTelemetry, ensuring data privacy by systematically dropping sensitive message content at the parse level.
- Architected a secure PostgreSQL control plane with row-level security to ingest evidence and issue DSSE-enveloped compliance packs signed via AWS KMS.

Rolestash — Local-First Chrome Extension & Job Tracker
Project Link: `github.com/SHINO-01/rolestash-extension`
Website Link: `rolestash.com`

- Architected, monetized, and shipped a local-first Kanban application to the Chrome Web Store, featuring cross-device synchronization and a secure payment gateway integration.
- Implemented complex DOM-parsing algorithms to extract job metadata from diverse career portals and seamlessly auto-populate application forms.
- Engineered a backend integration capable of parsing forwarded application emails (e.g., interview invites, rejections) via regex pattern matching to automatically update and transition Kanban card statuses.

NetZero Hub — Microservice Architecture for Carbon Tracking
Project Link: `github.com/SHINO-01/netZero`
Web-app Link: `netzero-zeta.vercel.app`

- Used TypeScript, Express and Docker Compose to build an API gateway and five independently deployable microservices, each with its own SQLite database, which delivered EV charging, journey planning and carbon-offset purchasing through one citizen web app, with double-bookings blocked by locked database transactions.
- Used RabbitMQ with publisher confirms, an outbox, auto-reconnect and idempotent consumers to make EV billing and offset certification asynchronous, which kept charging sessions completing during a billing-service or broker outage, with every receipt delivered automatically once the service came back.
- Used pm4py, Disco and Apromore to mine 4,081 events from 371 simulated cases and check them against the original BPMN process designs, which achieved 0.94–1.00 fitness and 0.92–1.00 precision, and identified 23 cases where real behaviour departed from the design.